Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/19/2025
PI Name:	Fuller, Charles A. Ph.D.		
Project Title:	Adaptation of Mouse Systems Physiology to Artificial Gravity via Centripetal Acceleration: Timing, Metabolism, and Aging		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	95616-5270	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	08/04/2020	**End Date:**	04/06/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2026 per F. Hernandez/ARC (Ed., 3/10/26). NOTE: End date changed to 04/06/2026 per F. Hernandez/ARC (Ed., 10/7/25). NOTE: End date changed to 09/03/2025 per F. Hernandez/ARC (Ed., 10/8/24). NOTE: End date changed to 09/03/2024 per NSSC information (Ed., 5/17/24). NOTE: End date changed to 05/03/2024 per NSSC information (Ed., 8/18/23).		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Hoban-Higgins, Tana Ph.D. (University of California, Davis) Fuller, Patrick M. Ph.D. (Unitersity of California, Davis)		
Grant/Contract No.:	80NSSC20K1501		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This program was designed to elucidate the dose-response curve for an artificial gravity (AG) exposure countermeasure using an established mouse model and centrifuge aboard the ISS. We proposed to record circadian rhythms of body temperature, food intake, rest/activity, and other video-recorded behaviors during four weeks of AG exposure produced via centrifugation in up to four sequential ISS studies as well as to examine the effects of AG on clock genes underlying circadian function. Additional outcome measures using 'omics analyses (e.g., proteomics and transcriptomics) on select tissues postflight would be used to examine regulatory factors and biomarkers related to our physiological and behavioral measures. Data from subjects exposed to AG will be compared with cohorts studied in microgravity and terrestrial ground controls. This series of studies would be the first to establish a dose-response curve for an AG countermeasure and to answer whether such countermeasures would also be necessary during lunar or Martian habitation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exposure to the spaceflight environment adversely affects many aspects of physiology and behavior. Among these deleterious changes are circadian and sleep/wake function alterations, which exacerbate physiological and performance decrements and link to disrupted metabolism and elevated aging biomarkers. Artificial gravity (produced via the centripetal acceleration of centrifugation) has long been proposed as a countermeasure to mitigate or prevent these deleterious changes during spaceflight. It has proved successful in a limited study on mice in the areas of muscle and bone. Planning for a mission to Mars raises the additional question of how effective the Martian gravitational environment will be in combatting microgravity-induced deficits. Finally, for an AG countermeasure to be successfully implemented, the parameters of AG exposure must be experimentally defined. This study will be the first to establish a dose-response curve for an AG countermeasure and to answer whether such countermeasures would also be necessary during lunar or Martian habitation.
Task Progress:	Much of the work accomplished during this time has been devoted to the analysis of the inflight video recordings and the tissues collected following the spaceflight experiment. Four groups of mice served as flight subjects of this study, each of which was exposed to a different level of g produced via centrifugation on the International Space Station using the Japanese Space Agency (JAXA) Cell Biology Experiment Facility (CBEF) facility. The centrifuges produced the following ambient force environments: 0G (microgravity spaceflight environment), 0.33G, 0.67G, and 1.0G. 1G control groups were studied concurrently on Earth. Following the successful conclusion of the spaceflight and ground-based experiments, requested tissues (brain, liver, vestibular apparatus) were collected, processed, and shipped for further analyses. Analysis procedures for these tissues were developed and tested. Proteomic and transcriptomic analyses are ongoing. Video recordings were collected at timepoints early, mid, and late during the mission to allow for analysis of behavioral data. At each of these times, video was recorded in 3-hour bins, 3 x per day (i.e., from 1100-1400, 1600-1900, and 2330-0230) for 2 successive days. Each three-hour block consisted of successive 30-minute videos from three subjects; thus, two 30-minute video recordings were collected from each subject during each 3-hour video window. Videos are being scored for observable behaviors. Following this, these results will be analyzed to elucidate the 24-hour pattern of behavior and to compare these between experimental groups. We will also compare results between experimental and ground control groups, as well as determine changes that may have occurred across the flight timeline.
Bibliography Type:	Description: (Last Updated: 12/08/2025)